Dvodimenzijska kontinuirana slučajna varijabla – nezavisna

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1 Funkcija (zajedničke) gustoće (vjerojatnosti)

$$f(x,y) = xy, \qquad 0 \le x \le 1,\ 0 \le y \le 2 \qquad (0 \text{ inače})$$

Provjera normiranosti (integral po cijeloj domeni mora biti jednak 1):

$$\int_0^1 \int_0^2 xy\,dy\,dx = \int_0^1 x\,dx \cdot \int_0^2 y\,dy = \frac{x^2}{2}\bigg|_0^1 \cdot \frac{y^2}{2}\bigg|_0^2 = \frac{1}{2} \cdot 2 = 1 \quad \checkmark$$

2 Marginalne funkcije gustoće (vjerojatnosti)

$$f(x) = \int_c^d f(x,y)\,dy$$

$$f(x) = \int_0^2 xy\,dy = x\int_0^2 y\,dy = x\,\frac{y^2}{2}\bigg|_0^2 = x\left(\frac{2^2}{2} - \frac{0^2}{2}\right) = x\,\frac{4}{2}$$
$$f(x) = 2x, \qquad 0 \le x \le 1$$

$$f(y) = \int_0^1 xy\,dx = y\int_0^1 x\,dx = y\,\frac{x^2}{2}\bigg|_0^1$$
$$f(y) = \frac{y}{2}, \qquad 0 \le y \le 2$$

$$f(x,y) = xy = 2x \cdot \frac{y}{2} = f(x)f(y) \longrightarrow \text{nezavisne varijable}$$

3 Uvjetne funkcije gustoće (vjerojatnosti)

$$f(x \mid y) = \frac{f(x,y)}{f(y)} = \frac{xy}{\frac{y}{2}} = 2x = f(x) \longrightarrow \text{nezavisne varijable}$$

$$f(y \mid x) = \frac{f(x,y)}{f(x)} = \frac{xy}{2x} = \frac{y}{2} = f(y) \longrightarrow \text{nezavisne varijable}$$

4 Matematička očekivanja

$$E(X) = \int_a^b x f(x)\,dx$$

$$E(X) = \int_0^1 x \cdot 2x\,dx = \int_0^1 2x^2\,dx = \left.\frac{2x^3}{3}\right|_0^1$$
$$E(X) = \frac{2}{3}$$

$$E(Y) = \int_0^2 y \cdot \frac{y}{2}\,dy = \int_0^2 \frac{y^2}{2}\,dy = \left.\frac{y^3}{6}\right|_0^2$$
$$E(Y) = \frac{4}{3}$$

5 Varijance

$$V(X) = \int_a^b \big(x - E(X)\big)^2 f(x)\,dx = \int_a^b x^2 f(x)\,dx - E(X)^2$$

$$V(X) = \int_0^1 x^2 \cdot 2x\,dx - \left(\frac{2}{3}\right)^2 = \int_0^1 2x^3\,dx - \frac{4}{9} = \left.\frac{2x^4}{4}\right|_0^1 - \frac{4}{9} = \frac{2}{4} - \frac{4}{9}$$
$$V(X) = \frac{1}{18}$$

$$V(Y) = \int_0^2 y^2 \cdot \frac{y}{2}\,dy - \left(\frac{4}{3}\right)^2 = \int_0^2 \frac{y^3}{2}\,dy - \frac{16}{9} = \left.\frac{y^4}{8}\right|_0^2 - \frac{16}{9} = \frac{16}{8} - \frac{16}{9}$$
$$V(Y) = \frac{2}{9}$$

6 Funkcije uvjetnog očekivanja (funkcije regresije)

$$E(X \mid y) = \int_a^b x f(x \mid y)\,dx$$

U našem slučaju je

$$f(x \mid y) = 2x = f(x) \;\longrightarrow\; \text{nezavisne varijable}$$

$$f(y \mid x) = \frac{y}{2} = f(y) \;\longrightarrow\; \text{nezavisne varijable}$$

pa je

$$E(X \mid y) = E(X) = \frac{2}{3}, \qquad E(Y \mid x) = E(Y) = \frac{4}{3}.$$

7 Kovarijanca

$$\mathrm{Cov}(X,Y) = \int_a^b \int_c^d \big(x - E(X)\big)\big(y - E(Y)\big) f(x,y)\, dy\, dx$$

ili (jednostavnije):

$$\mathrm{Cov}(X,Y) = \int_a^b \int_c^d xy\, f(x,y)\, dy\, dx - E(X)E(Y)$$

$$\begin{aligned}
\mathrm{Cov}(X,Y) &= \int_0^1 \int_0^2 xy \cdot xy\, dy\, dx - \frac{2}{3} \cdot \frac{4}{3} = \int_0^1 \int_0^2 x^2 y^2\, dy\, dx - \frac{8}{9} \\
&= \frac{x^3}{3}\Big|_0^1 \cdot \frac{y^3}{3}\Big|_0^2 - \frac{8}{9} = \frac{1}{3} \cdot \frac{8}{3} - \frac{8}{9} \\
\mathrm{Cov}(X,Y) &= 0
\end{aligned}$$

8 Koeficijent korelacije

$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma(X)\,\sigma(Y)} = \frac{0}{\sigma(X)\,\sigma(Y)} = 0$$

Napomena: Ako su slučajne varijable nezavisne, kovarijanca je uvijek nula, ali ima teoretskih razdioba kod kojih je kovarijanca nula, a slučajne varijable nisu nezavisne!